

High-Intent Propensity & Behavioral Intelligence

High-Intent Behavioral Intelligence — Bumble Dates Experience This dashboard operationalizes a reproducible measurement pipeline that surfaces who on the platform is most likely to progress from browsing to meaningful connections. Built from a Single Source of Truth in BigQuery and a Colab modeling workflow, it combines behavioral clustering, propensity scoring, and SHAP interpretability to translate product signals (sessions, matches, match rate, verification, and Dates usage) into decision-ready insights for Product, CRM, Safety, and Commercial stakeholders; the work is framed to support Bumble's quality-first strategy, causal evaluation of the Dates rollout, and the DRI Data Scientist responsibilities described in the role research.



Executive Overview



Total Users

50,000

High-Intent Share

4.03%

Verification Rate

92.09%

Avg Sessions (30d)

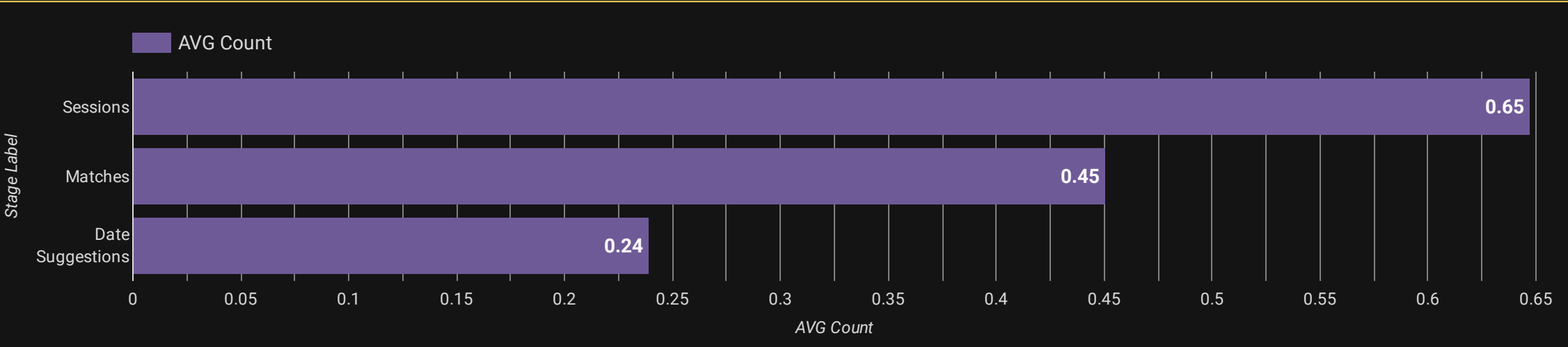
0.65

Avg Matches (30d)

0.45

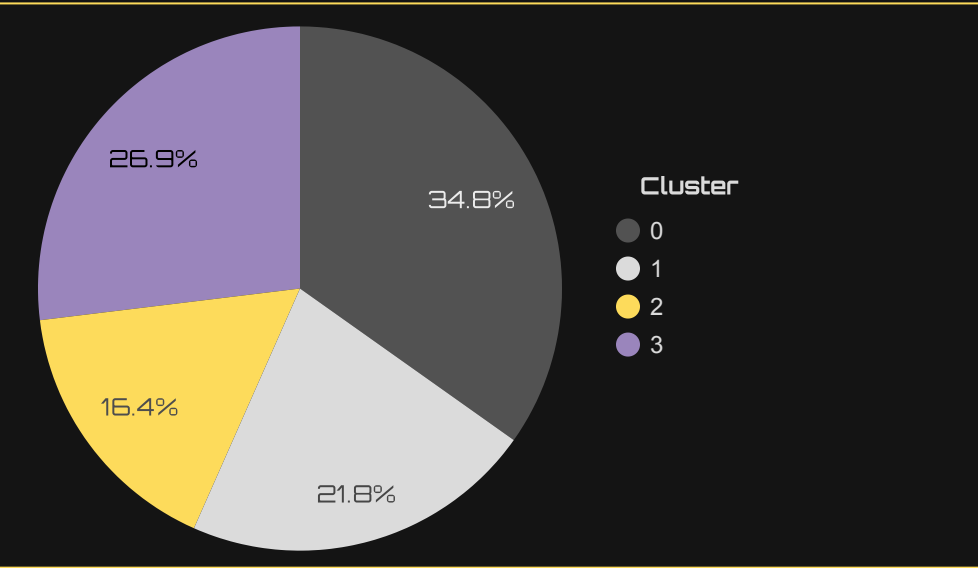
Engagement Funnel (sessions → matches → dates)

Highlights where users drop off between browsing, matching, and progressing to dates, pointing to the stages most in need of UX or feature improvements.



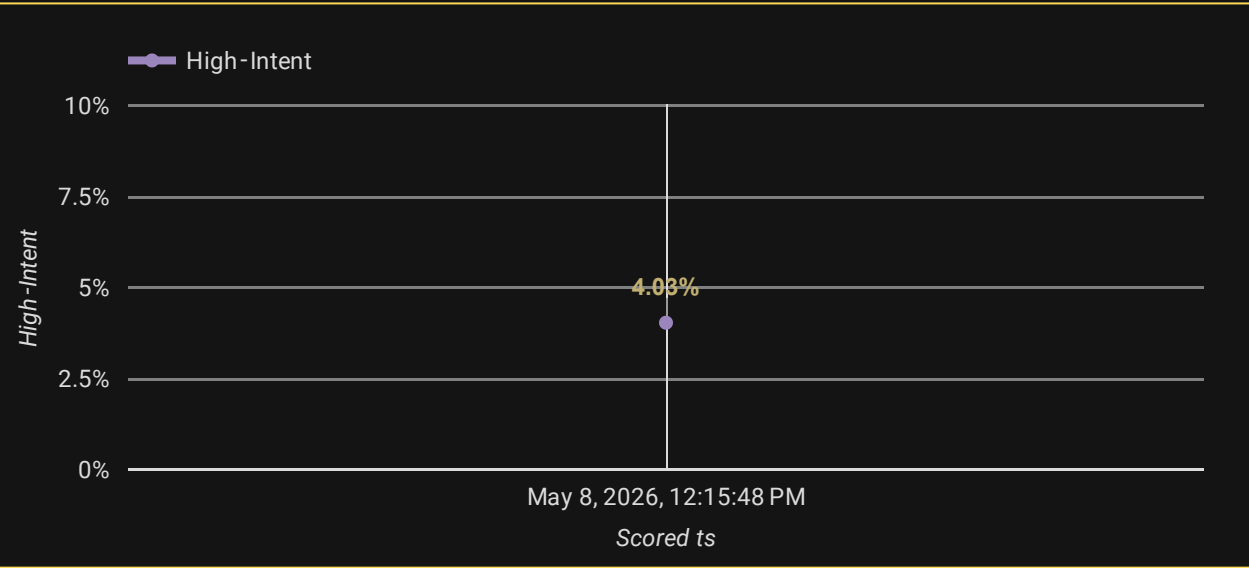
Cluster Distribution

Shows how users are distributed across behavioral segments, helping prioritize which groups to focus on for growth or intervention.

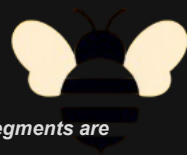


High-Intent Trend Over Time

Tracks whether high-intent behavior is increasing or declining, linking product and marketing changes to shifts in user quality.

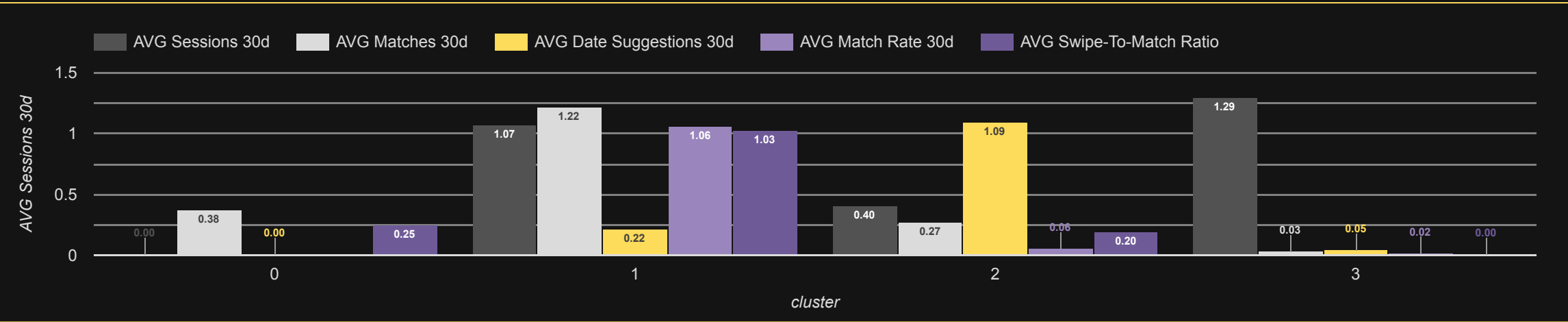


Behavioral Segmentation



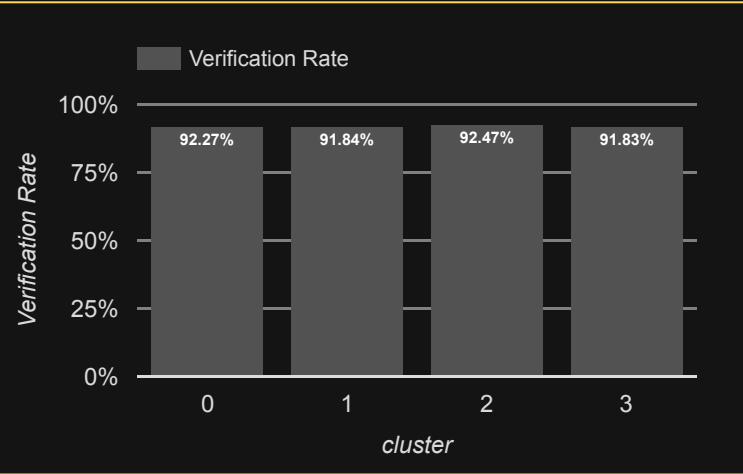
Cluster Profiles (Mean Feature Values)

Compares core behaviors across clusters, revealing which segments are most engaged, most successful, or most at risk of churn.



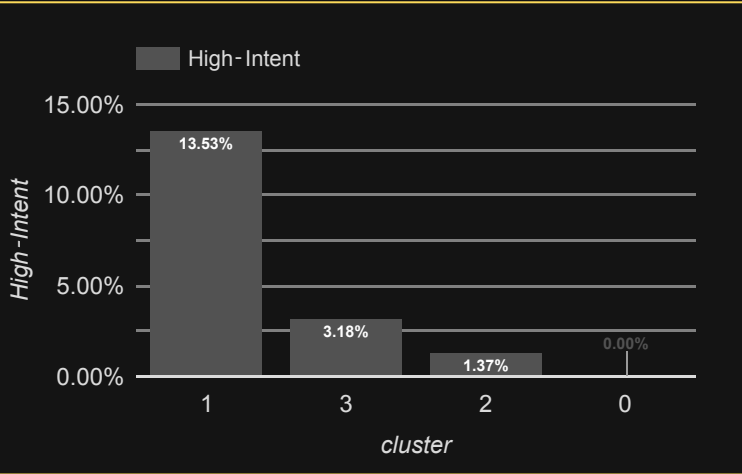
Shows how safety adoption varies by segment, indicating where verification nudges or education may be most needed.

Cluster-Level Verification Rate



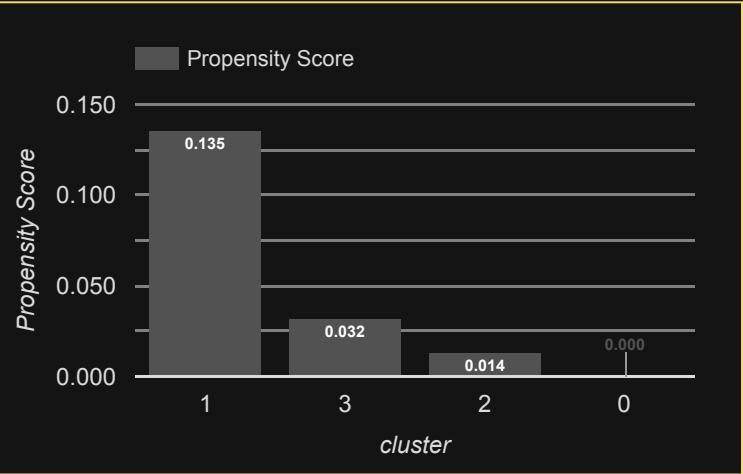
Identifies which clusters contain the highest concentration of high-intent users, guiding targeting for premium features or tailored experiences.

Cluster-Level High-Intent Share



Summarizes how likely each cluster is to contain high-intent users, helping prioritize segments for product investment and experimentation.

Cluster-Level Propensity Score Distribution



Top 10% Users by Propensity Score Within Each Cluster

Surfaces the most promising users in every segment, enabling targeted outreach, white-glove experiences, or early adopter programs.

Cluster		User ID	Propensity Score ▾
1.	1	user_014191	0.99935204
2.	1	user_008638	0.99935204
3.	1	user_021070	0.99935204
4.	1	user_039056	0.99935204
5.	1	user_013419	0.99935204

High-Intent Propensity

High-Intent Users (Count)

2,017

Quantifies the absolute pool of high-intent users available for premium experiences, targeted campaigns, or strategic focus.

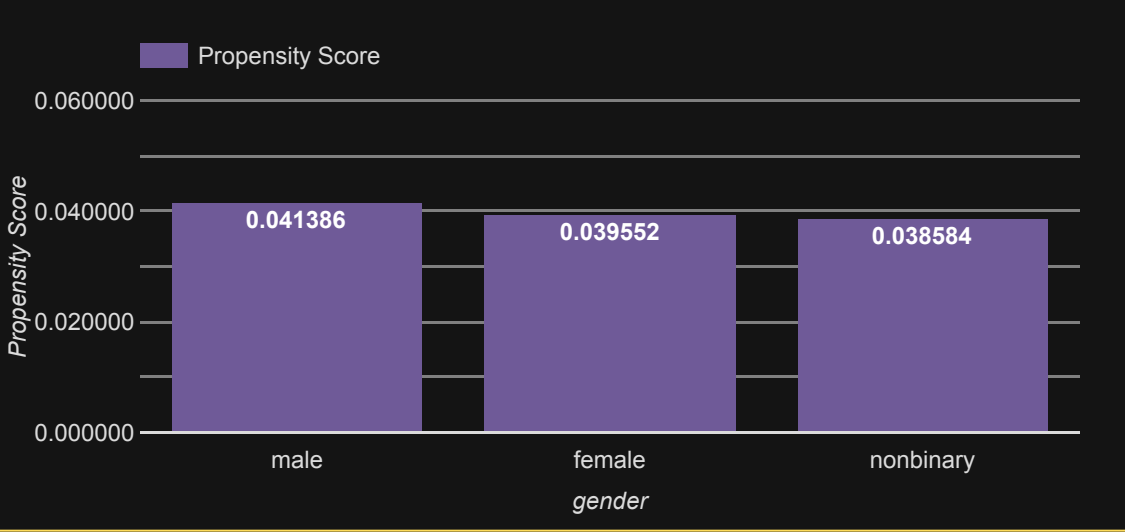
Shows how concentrated or diffuse high-intent likelihood is across the user base, revealing whether high-intent is rare, emerging, or widespread.

Propensity Score Histogram



Compares predicted high-intent across genders, helping detect potential imbalances that may warrant product, marketing, or fairness review.

Propensity Score by Gender



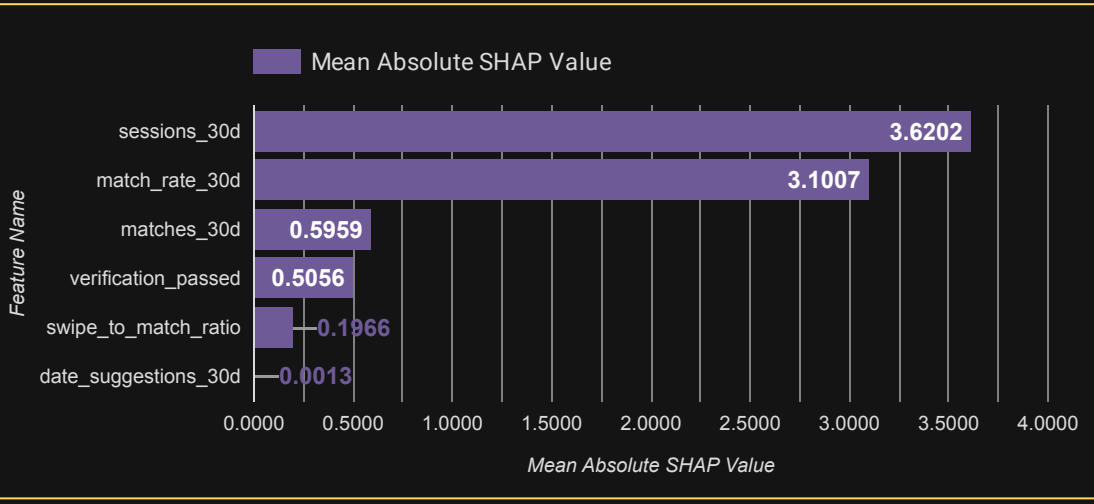
Avg Propensity Score

0.04

Summarizes the overall “intent health” of the user base, indicating whether the platform is trending toward more serious, engaged usage.

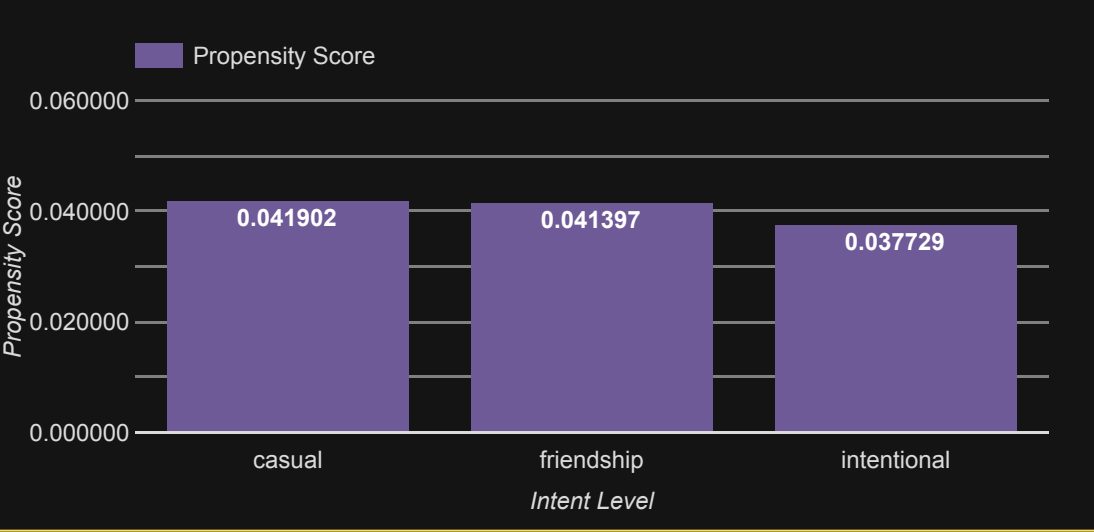
Ranks which behaviors and signals most influence the model’s predictions, translating a complex model into an actionable list of levers the business can pull.

SHAP Summary Plot



Tests whether stated intent aligns with modeled behavior, highlighting gaps between what users say they want and how they actually behave.

Propensity Score by Intent Level

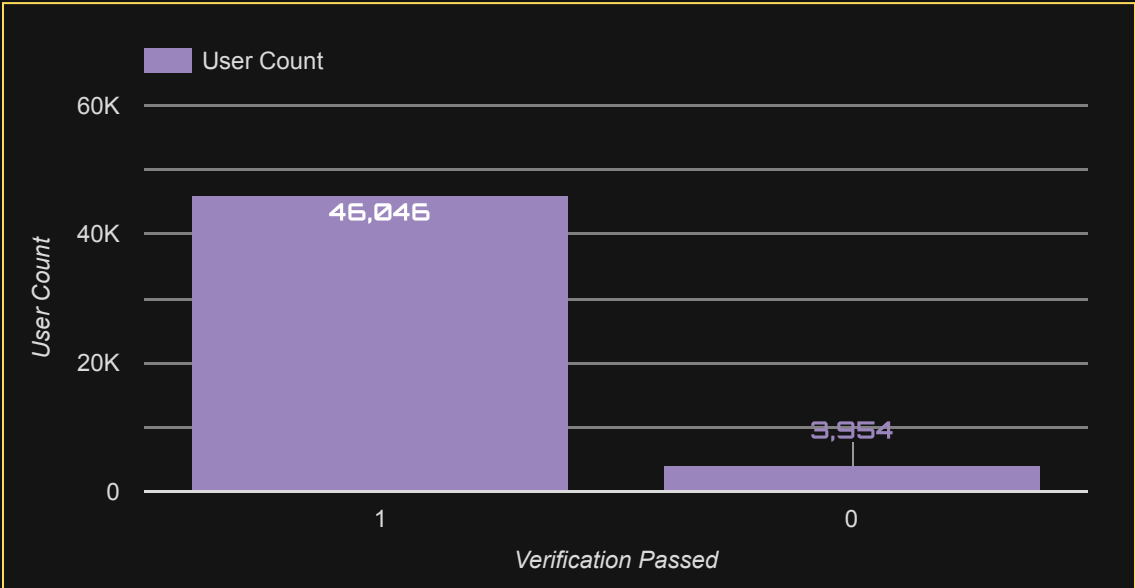


Safety & Verification



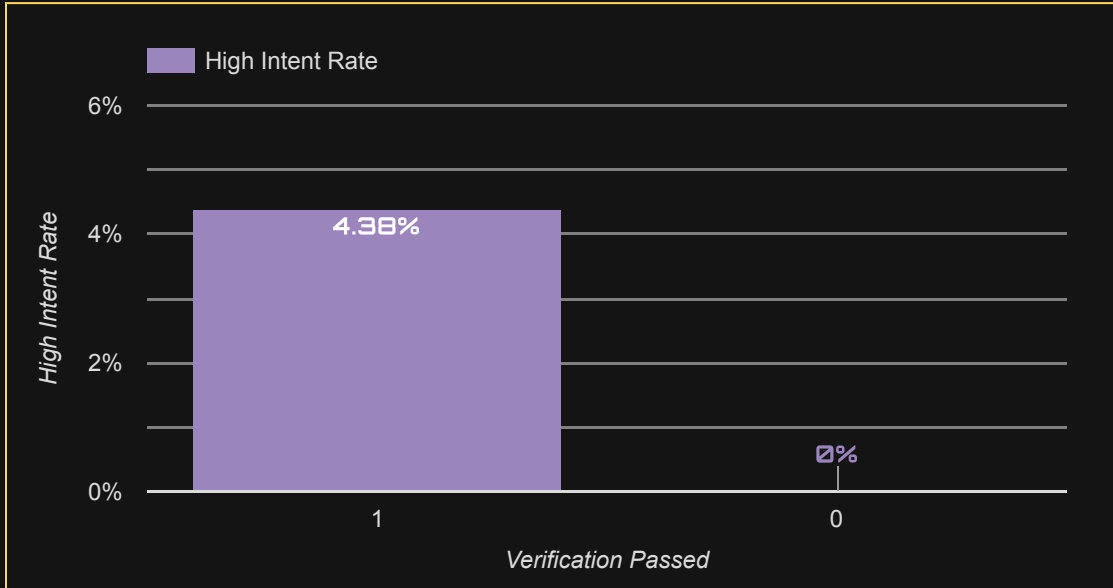
Shows how many users are verified versus unverified, indicating the strength of the safety foundation at any point in time.

Verification Status Distribution



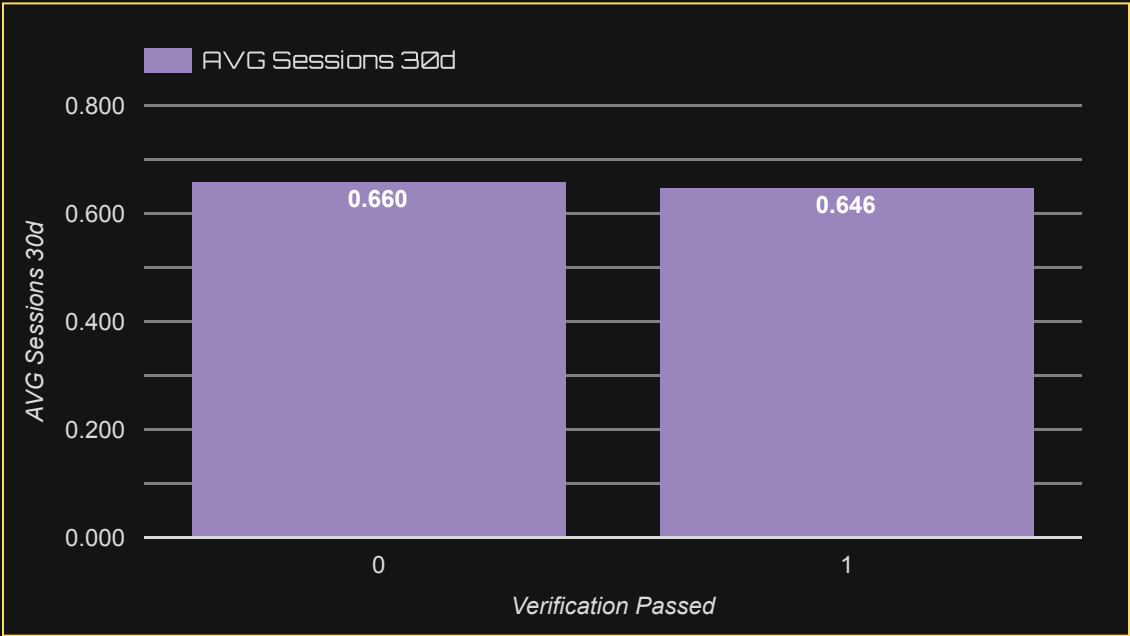
Compares high-intent rates by verification status, revealing how strongly safety and seriousness are linked on the platform.

High-Intent vs. Unverified Comparison



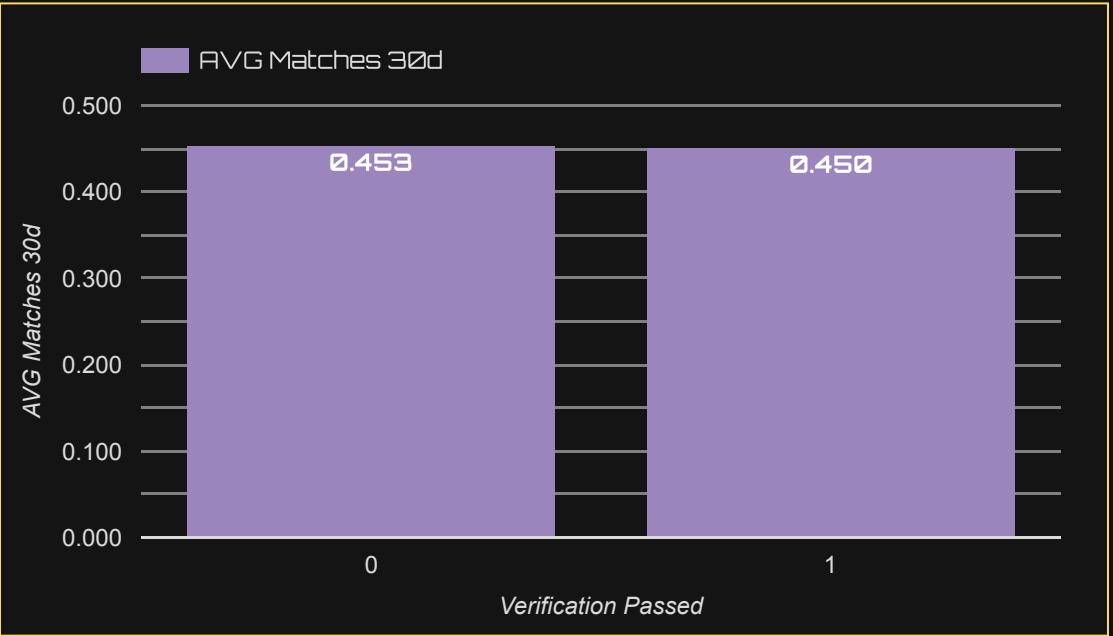
Examines whether verified users engage more frequently, informing whether verification also correlates with healthier usage patterns.

Avg Sessions (30d) by Verification Status



Highlights whether verified users achieve more matches, supporting decisions around promoting verification as a way to improve outcomes.

Avg Matches (30d) by Verification Status

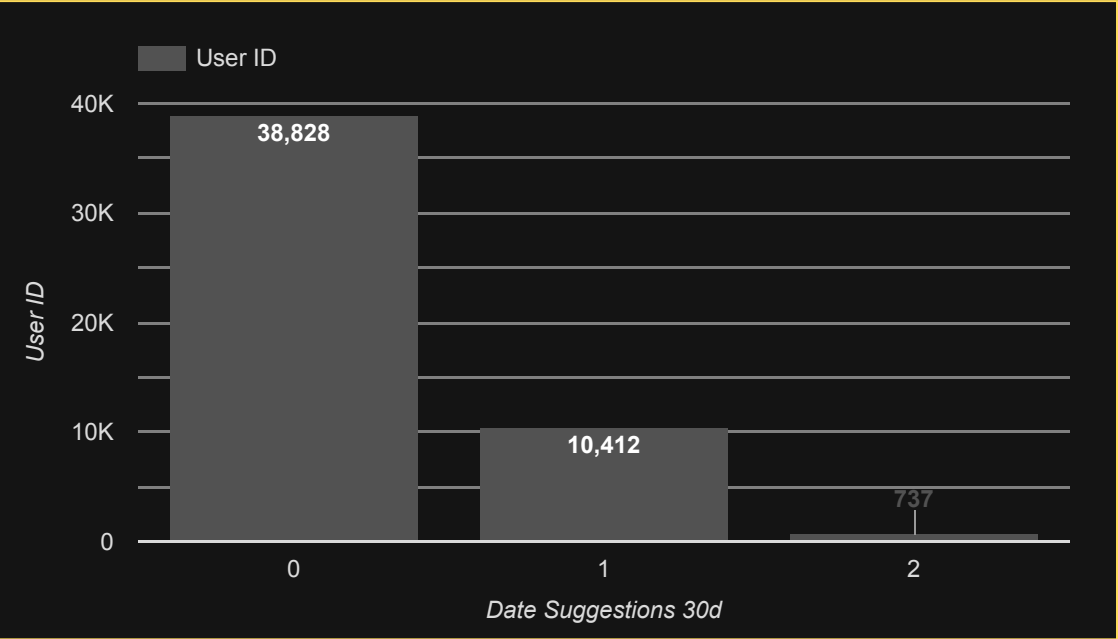


Dates Feature Insights



Shows how many users actually use the Dates feature, revealing adoption gaps and the size of the opportunity space.

Date Suggestions (30d) Distribution



Profiles the users who lean into Dates, helping product and CRM teams design experiences tailored to this high-potential segment.

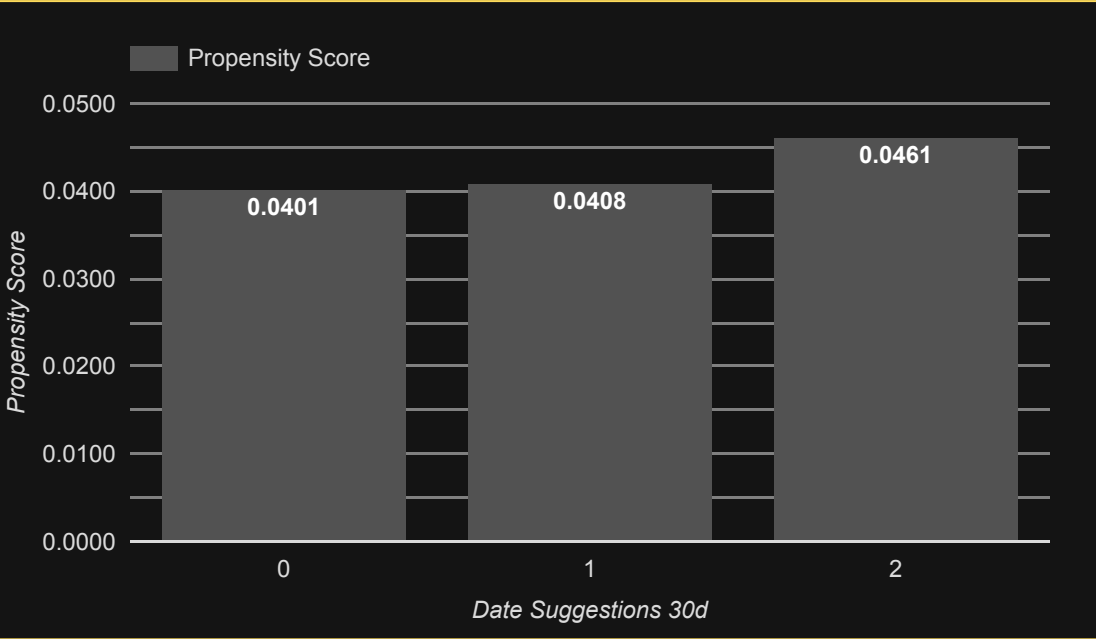
Cluster 2 – Dates-Forward Users

	User ID	Sessions 30d	Matches 30d	Date Suggestions 30d	Propensity Score
1.	user_04563_9	2	1	2	0.999245
2.	user_047814	2	1	2	0.999245
3.	user_002614	2	1	2	0.999245
4.	user_016482	2	1	2	0.999245
5.	user_00601_3	2	1	2	0.999245
6.	user_03613_5	2	1	3	0.999245
7.	user_011021	2	1	2	0.999245
8.	user_02644_2	2	1	2	0.999245
9.	user_017628	2	1	2	0.999245
10.	user_04920_9	2	1	2	0.999245
11.	user_03345_7	2	1	2	0.999245

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Links Dates usage to modeled high-intent, indicating whether leaning into the feature is associated with more serious, committed behavior.

High-Intent Propensity vs. Date Suggestions



Reveals how efficiently Dates-forward users move from activity to matches to dates, pinpointing where improvements could unlock more real-world connections.

Cluster 2 Engagement Funnel

